

ADAM MLAYEL

Software Developer | Python | AI & Data Science

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LinkedIn

Portfolio

Tunis, Tunisia

PROFESSIONAL SUMMARY

First-year engineering student passionate about Artificial Intelligence, Data Science, and software development. Strong interest in Python-based automation and intelligent systems, with hands-on experience through academic and personal projects involving AI models, backend development, and system design.

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, C, C#, Dart
- AI/ML:** PyTorch, TensorFlow, Keras, Scikit-learn
- Python:** NumPy, Pandas, Matplotlib
- Web:** HTML, CSS, FastAPI
- Mobile/Game:** Flutter, Unity
- Tools:** Git, Docker, Anaconda
- Databases:** PostgreSQL, MySQL, Firebase
- Design:** Figma, Photoshop, Blender
- OS:** Windows, Ubuntu, Kali Linux

EDUCATION

Bachelor's Degree in Computer Science—Software Engineering and Information Systems	Sep 2022 – Jun 2025
<i>Higher Institute of Computer Science and Mathematics of Monastir</i>	<i>Monastir, Tunisia</i>
Baccalaureate in Experimental Sciences	Sep 2020 – Jun 2021
<i>Ennacer Private School</i>	<i>Gafsa, Tunisia</i>

PROFESSIONAL EXPERIENCE

Manager Partner Software (MpSoft)	Aug 2024
<i>Summer Intern – AI System for System Log Analysis</i>	<i>Les berges du lac, Tunisia</i>
- Conducted needs analysis and defined functional specifications for AI system analyzing maintenance logs for preventive maintenance. - Designed architecture and prepared technical environment for automated error detection module from log files. - Developed conceptual prototype and detailed plan for automated anomaly reporting system.	
Gafsa Phosphate Company	Jun 2024
<i>Summer Intern – Desktop Application Development</i>	<i>Gafsa, Tunisia</i>
- Designed and developed desktop application for Human Resources data management. - Developed modules for automated calculation of HR statistical variances.	

KEY PROJECTS

Web Application: Electro Repair Platform	Jan 2026
- Developed a full-stack web application allowing users to submit and track electronic repair requests in real time. - Built role-based workflows for users, technicians, and administrators, including ticket assignment, status tracking, and dashboards. - Tech Stack: Next.js (App Router, TypeScript, Tailwind CSS), FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT Authentication, Docker, RESTful APIs	
Coffre-Fort Documentaire Sûr et Intelligent	2025
<i>Containerized microservices-based document management system with AI, OCR, and SSO</i>	
- Designed and deployed a secure, containerized document management platform using a microservices architecture. - Implemented OCR-based text extraction and AI-powered document summarization (French) with keyword generation using Ollama and GPU acceleration. - Integrated Keycloak OIDC single sign-on with granular, time-limited access control (view, download, OCR, AI, upload permissions). - Tech Stack: React 18, Node.js (Express), Mayan EDMS (Django), Ollama (LLM), Keycloak, PostgreSQL, Redis, Docker, NVIDIA GPU support.	
Final Year Project: Intelligent Solution for Agriculture and Plant Health	Feb 2025 – Jun 2025
<i>Multilingual Mobile App (FR/EN/AR) with AI Agent for Plant Identification and Health Diagnosis</i>	
- Developing custom-trained MobileNetV2 Deep Learning model for plant identification and health prediction from images. - Integrating generative AI for personalized care recommendations based on weather data and plant condition. - Tech Stack: Flutter (frontend), FastAPI (backend), Firebase (authentication & real-time messaging).	

LEADERSHIP & COMMUNITY EXPERIENCE

Media Manager	2024 – Present
<i>Gaming Lab Club, TekUp University</i>	<i>Tunis, Tunisia</i>
- Manage the club's media presence, including social media content creation, visual design, and event promotion. - Collaborate with technical teams to showcase gaming, development, and innovation activities through digital platforms.	
University Ambassador	2022
<i>University of Monastir</i>	<i>Monastir, Tunisia</i>
Invited to support the organization of a three-day hackathon, contributing to event coordination in a professional academic environment.	
Vice President	2016 – 2020
<i>Von Club</i>	<i>Gafsa, Tunisia</i>
- Progressed from active member to Vice President in an anime and gaming event organization club. - Developed leadership, event management, and community engagement skills through team coordination and large-scale event planning.	

CORE COMPETENCIES

Problem Solving • Creative Thinking • Adaptability • Initiative • Technical Communication

LANGUAGES

Arabic (Native) English (Professional) French (Professional) German (Beginner)